

Early childhood education and early childhood special education teachers' perceived stress, burnout, and depressive symptoms

Hyun-Joo Jeon^{a,*}, Kyong-Ah Kwon^b, Christina McCartney^c, Lindsay Diamond^c

^a Human Development, Family Science, & Counseling, College of Education & Human Development, University of Nevada, Reno, 1664 N. Virginia St., Reno, NV, USA

^b Department of Instructional Leadership and Academic Curriculum, Jeannine Rainbolt College of Education, University of Oklahoma, Tulsa, OK, USA

^c Special Education, College of Education & Human Development, University of Nevada, Reno, NV, USA

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ABSTRACT

The current study examines the relationships between teachers' job demands and resources, perceived stress, job burnout, and depressive symptoms through an online survey administered to early childhood education (ECE: $n = 143$) and early childhood special education (ECSE: $n = 121$) teachers in a western state of the United States. Results showed no difference between ECE and ECSE teachers in their levels of job demands, resources, perceived stress, job burnout, and depressive symptoms. Both ECE and ECSE teachers' job-related stressors were positively related to their perceived stress, and their job commitment was negatively associated with job burnout. However, there were some differences in the correlates of job burnout between ECE and ECSE teachers. For ECE teachers, the number of children with disabilities and their teaching efficacy predicted their job burnout, whereas with ECSE teachers, job-related stressors and the supportiveness of school climate (or the lack thereof) were indirectly related to their job burnout through their perceived stress. The current study suggests that providing professional development for these teachers and enhancing a school's supportive climate will promote the well-being of both ECE and ECSE teachers.

1. Introduction

Early childhood education (ECE) and early childhood special education (ECSE) teachers' job burnout and depressive symptoms have been recognized as an important issue (Kwon et al., 2020; Carver-Thomas & Darling-Hammond, 2017; Johnson et al., 2021; Li et al., 2020; Linnan et al., 2017; Robinson et al., 2019; Schaack et al., 2020). Researchers have documented that teachers' job burnout and depressive symptoms in various ECE and ECSE settings are related not only to rates of teacher turnover (Kwon et al., 2020; Kwon et al., 2021) and the quality of teaching in their classroom (Kwon et al., 2019) but also to their students' behavioral problems (Kwon et al., 2019; Hamre & Pianta, 2004; Jeon et al., 2014; Roberts et al., 2019) and Individualized Education Program (IEP) outcomes (Wong et al., 2017). Given the potential impact of job burnout and depressive symptoms of ECE and ECSE teachers on their teaching practice and child outcomes, it is crucial to explore what job-related factors are associated with their well-being: such as perceived stress, job burnout, and depressive symptoms. Our study examined these factors and their associations that will have implications for program and system-level changes and interventions designed to improve these

teachers' well-being and the quality of their teaching.

The Job Demand-Resource (JD-R) model (Bakker & Demerouti, 2017), a widely used framework across occupational settings, explains teachers' job-related and psychological well-being associated with their job demands and resources (job resources and internal resources). This model proposes two distinct mechanisms through which job demands and resources are associated with various psychological and professional outcomes (Bakker & Demerouti, 2017; Hakanen et al., 2006). We used the JD-R model in the current study to examine the role of job demands and available resources (i.e., job resources, as well as internal resources) on several aspects of the well-being of ECE and ECSE teachers. We also tested mediational relationships by which job-related factors are associated with teachers' perceived stress, which are, in turn, related to their job burnout and depressive symptoms.

1.1. Working conditions and Well-Being of ECE and ECSE teachers

Both ECE and ECSE teachers are often identified as groups of teachers who are likely to experience job-related stress and job burnout (Vandenberghe & Huberman, 1999) and have a high rate of job turnover

* Corresponding author.

E-mail addresses: hyunjooj@unr.edu (H.-J. Jeon), kkwon@ou.edu (K.-A. Kwon), cmccartney@unr.edu (C. McCartney), ldiamond@unr.edu (L. Diamond).

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(Miller & Bogatova, 2009; Wells, 2015). More than a decade ago, educational reforms were made in both the fields of ECE and ECSE teaching which required that there be higher standards for both teachers in those fields and for their programs; these reforms came along with an increase in the number of children these teachers needed to serve in their classroom (Bullough et al., 2012; Sacco, 2011). For example, Head Start reauthorization in 2007 mandated increased teacher educational qualifications and higher standards for classroom quality (Bullough et al., 2012). These requirements resulted in an increase in job demands (e.g., extra paperwork and additional assessments) on these teachers, which also increased the stress on these teachers above and beyond the already existing challenges of working with children and families of many diverse backgrounds who often also live in poverty (Kwon et al., 2021; Bullough et al., 2012; Li-Grining et al., 2010; West et al., 2018; Zhai et al., 2011). For ECSE teachers, the *Individuals with Disabilities Education Improvement Act of 2004* (IDEA, 2004) mandates that states provide a free and appropriate public education (FAPE) to young children with disabilities in the least restrictive environment (LRE) possible. Since the reauthorization of IDEA, the expanded eligibility for services has meant that the number of children receiving special education services has grown (Sacco, 2011), and has been accompanied by an increase in parents' expectations of what their child will receive, all of which has added more job demands on ECSE teachers.

In general, ECE and ECSE teachers may experience similar job demands and resources, as they all work in educational settings with the same age group of children of diverse backgrounds and abilities. Piper (2007) and Stayton and Bruder (1999) have argued that there are more similarities than differences between the teachers' work in these two fields. However, it is also plausible that there may be differences between ECE and ECSE teachers in the associations between their job demands, resources, and well-being. These two workforces have major differences in their required job qualifications and training (e.g., different educational levels, ranging from high school diplomas for ECE teachers to bachelor's degree requirement for ECSE teachers). These teachers also have varying levels of job demands and resources due to different program or school organizations and the range of their students' abilities, which may be differently related to teachers' well-being. ECE teachers may experience more demands and stress, as they typically have more children to serve than ECSE teachers (Sacco, 2011). Also, teaching in ECE has been known to be one of the lowest paid professions and to be particularly stressful (Johnson et al., 2021). Conversely, ECSE teachers may feel more stress when working with children with disabilities, which may be more physically and emotionally demanding than the work of ECE teachers. Thus, it is important to acknowledge potential differences in their job demands and resources and to better understand how they play a role in teachers' well-being and to be better able to provide these teachers with tailored support and resources. However, there is no study, to the authors' knowledge, that has examined both ECE and ECSE teachers' job demands and resources and their associations with well-being. More research is needed to understand these two groups of teachers' job-related experiences and their well-being.

1.2. Association of job demands with perceived Stress, Burnout, and depression

Job demands are aspects of one's work that require sustained physical or mental effort (Demerouti et al., 2001). In this study, we included two types of job demands specific to ECE and ECSE teachers: job-related stressors and the number of children in a classroom with disabilities. We examined how these are associated with ECE and ECSE teachers' perceived stress, job burnout, and depressive symptoms.

1.3. Job-related stressors

Among job demands, job-related stressors are teacher's perception of

a threat to their well-being from job demands and work conditions, which may cause unpleasant emotions (Kyriacou, 2001; Kyriacou & Sutcliffe, 1978; Paquette & Rieg, 2017). Studies reveal that ECE teachers face stressors related to the high demands placed upon them, a lack of adequate support from their administrators and the inability to be autonomous in their classrooms; also, there are limited opportunities for their own professional development and training which could improve their working conditions, as well as there being high rates of teacher turnover, lack of respect for their profession, and low wages (Kwon et al., 2020; Cooper & Kelly, 1993; Jeon et al., 2018; Kyriacou, 2001; Smith & Lawrence, 2019; Whitebook & Sakai, 2003). The extant research related to ECSE teachers' job-related stressors has also identified a unique combination of stressors that include having to prepare IEPs and provide individualized instruction (Kwon et al., 2021), an abundance of paperwork (Male & May 1997), and inconsistent assistance from support staff (Hamama et al., 2013).

In several studies, these job-related stressors and the resulting perceived stress experienced by teachers have been identified as links to job burnout in teachers (Bakker et al., 2005; Hakanen et al., 2006; Yu et al., 2015) including ECSE teachers (Biglan et al., 2013) and ECE teachers (Aboagye et al., 2018; Farewell et al., 2022). ECE teachers with substantial job stressors may also experience higher rates of psychological distress, including perceiving being over-stressed and having depressive symptoms (Kwon et al., 2021). In fact, ECE teachers who perceived their work conditions as being stressful tended to experience more stress, experience more job burnout, and to have more depressive symptoms (Jeon et al., 2018). In a longitudinal study, Schonfeld (2001) found that both ongoing and episodic job stressors early in the school year were related to teachers' depressive symptoms toward the end of the school year.

1.4. Number of children with disabilities

Another important job demand to consider in examining teachers' well-being is the number of children with disabilities in both ECE and ECSE settings. In the United States, about 67 percent of the children between the ages of 3 and 5 who qualified for special education services under the IDEA received some special education services in general ECE classrooms (U.S. Department of Education, 2020). Among these children with disabilities, 40 % receive at least 10 h of special education service in general ECE classrooms (U.S. Department of Education, 2020). While working with children with diverse developmental needs is inevitable and can be rewarding for teachers, doing so may also serve as an additional job demand and strain for some teachers (Guglielmi & Tatrow, 1998), which may affect teachers' psychological well-being. This may particularly be the case for ECE teachers, who often may not have sufficient training or resources to work with children who have a variety of developmental needs and disabilities (Kwon et al., 2019; Cook, 2002; Shippen et al., 2005; Swain et al., 2012). Regardless, there is, to date, no study on the associations between the number of children with disabilities served by either ECE or ECSE teachers and these teachers' perceived stress, job burnout, and depression.

1.5. Association of Job-Related and internal resources with perceived Stress, Burnout, and depression

Consistent with the JD-R model, research has found that there are multiple job-related and individual internal factors related to teachers' well-being (Holzberger et al., 2021). Sufficient job resources consist of both physical and psychological working conditions which also help teachers achieve their professional goals and help them grow in their work (Bakker & Demerouti, 2017). In addition, teachers' professional internal resources can also contribute to their well-being and the quality of their teaching (Bakker & Demerouti, 2017). Although teachers may be exposed to relatively comparable working conditions and may experience similar job stressors, not all teachers experience similar

levels of psychological distress and job burnout (McLean et al., 2019). Teachers' professional internal resources, such as job commitment and teaching efficacy, may play a role in reducing or countering the possible adverse effects of job-related stressors and the lack of sufficient job support on teachers' well-being when they face extensive job demands. The present study included a positive school climate as a job resource and teachers' job commitment and self-efficacy as a professional internal resource for their well-being.

1.6. School climate as a job resource

For teachers, one of the most important job resources is a collaborative and supportive school climate, including instructional support, social relationships with colleagues (Richards et al., 2018; Shackleton et al., 2019; Skaalvik & Skaalvik, 2018), plus a degree of autonomy and control over instructional decisions (Collie et al., 2011; Loukas & Murphy, 2007). Teachers' perceptions of their school climate are negatively associated with their levels of perceived psychological stress (Bakker et al., 2005; Cappe et al., 2017) and job burnout (Cappe et al., 2017; Richards et al., 2018). Researchers have also found that teachers' perceptions about their workplace environment, which includes school climate, are linked to their depressive symptoms (e.g., Papastyliaou et al., 2009; Schonfeld & Bianchi, 2016; Shen et al., 2014). According to Maslach et al. (2001), both job burnout and depressive symptoms are associated with stressful environments; however, job burnout symptoms are caused by specific workplace environments, while depressive symptoms generally result from general negative life events (Maslach et al. 2001). Thus, school climate needs to be considered when examining teachers' well-being (Ahola et al., 2014; Bianchi et al., 2015; Maslach et al. 2001; Schonfeld & Bianchi, 2016).

1.7. Job commitment as an internal resource

One internal resource that may be associated with teachers' psychological distress is their job commitment – defined as a positive outlook and intrinsic motivation about their profession (Blau, 1985; Hackett et al., 2001; Landy & Guion, 1970). Teachers' commitment to their profession is positively associated with their performance and achievement (McInerney et al., 2015). On the other hand, studies have documented that ECE teachers' job commitment is negatively related to their perception of job stress (e.g., Jeon et al., 2019; Jepson & Forrest, 2006) and job burnout (Nesje, 2017). From these studies, we assume that ECSE teachers' positive views about being teachers and their commitment to teaching may also be an important factor associated with their perceived stress, job burnout, and depressive symptoms. However, research in this area is limited.

1.8. Self-Efficacy as an internal resource

Rotter (1966) conceptualized teaching efficacy based on self-efficacy or self-appraisal of the person's capabilities in their work (Bandura, 1977; Tschannen-Moran & Woolfolk Hoy, 2001; Wang et al., 2015). Teachers who believe that they can positively influence student outcomes through their teaching, and express confidence in their ability to motivate students and manage their classrooms effectively (Capara et al., 2006), which may be linked to their psychological well-being. In a systematic literature review, Zee and Koomen (2016) found that teaching efficacy is related to teachers' job burnout and psychological stress. Several researchers have found negative associations between K-12 teachers' teaching efficacy and job burnout, suggesting that teachers having high teaching efficacy may be confident in their job and, thus, experience less job burnout (Brouwers & Tomic, 2000; Martin et al. 2012; Skaalvik & Skaalvik, 2007). Further, Jeon et al. (2018) found that ECE teachers' teaching efficacy was negatively related to both teachers' perceived stress and job burnout, noting that there was a need for future studies examining the relationship between teaching efficacy and

psychological well-being.

The role of teaching efficacy in teachers' depression has also been examined in the K-12 setting (e.g., Capone & Petrillo, 2020; Huang et al., 2019; Schwerdtfeger et al., 2008). For example, elementary school teacher's self-efficacy was negatively associated with their anxiety and depression in a study of teachers in Hong Kong (Huang et al., 2019). Italian high school teachers' self-efficacy was positively correlated with their emotional, psychological, and social well-being (Capone & Petrillo, 2020). Similarly, Schwerdtfeger et al. (2008) found that German school teacher's self-efficacy predicted their psychological well-being. However, the aforementioned studies on teaching efficacy were all conducted in general education settings with teachers of school-age children, which highlights the need to examine both ECE and ECSE settings.

1.9. Perceived stress as a potential mediator

Although perceived stress, job burnout, and depression share qualitative characteristics (Bianchi et al., 2016; Bianchi et al., 2016; Meier & Kim, 2022) and may co-develop (Iacovides et al., 2003), researchers have also suggested that these are separate conditions (Dell'Erba et al., 1994; Iacovides et al., 2003). For example, depression and job burnout may be caused by similar stressors (e.g., heavy workload, negative work environment) and perceived stress (Iacovides et al., 2003). However, job burnout has been tied mostly to the work environment, while depression is a condition not only associated with one's work, but also with other elements of one's life (Iacovides et al., 2003). Additionally, perceived stress may be more situational and momentarily accommodated (McGonigal, 2015), while job burnout is an ongoing negative emotional status caused by work environment, and symptoms of depression often demonstrate a more pervasive manifestation of overall emotional distress (American Psychiatric Association, 2013).

Furthermore, Schwarzer and Hallum (2008) found that perceived job stress mediated the link between teachers' self-efficacy and job burnout for K-12 teachers in Syria and Germany. That study suggests that teachers' internal resources are related to their well-being through perceived stress. Teachers' perceived stress is positively associated with job burnout (Jennings & Greenberg, 2009) and depressive symptoms (Whitaker et al., 2013). Teachers' perceived stress is manifested as an immediate reaction to high job-related demands and stressors, which could ultimately lead to job burnout and depressive symptoms. Thus, given the associations among perceived stress, job burnout, and depression implicated in the literature, it is reasonable to expect that perceived stress should play a mediating role in the association of job demands and resources with job burnout and depressive symptoms.

1.10. Current study

Based on the JD-R model, the current study examined whether ECE and ECSE teachers' job demands and resources predict their perceived stress, job burnout, and depressive symptoms by using a mediational model (see Fig. 1). As job demands, we included teachers' job-related stressors and the number of children with disabilities in their classrooms. In the category of job resources, we included a positive and supportive program/school climate and teachers' internal resources (job commitment and teaching efficacy). We also compared these relationships between ECE and ECSE teachers. We hypothesized that the pathways by which ECE and ECSE teachers' job demands and resources related to their job burnout and depression would differ even though other researchers have argued that there are both similarities and differences between the ECE and ECSE teachers' work (Piper, 2007; Stayton & Bruder, 1999). We argue that these teachers have different job qualifications and training and varying levels of job demands and resources due to various program or school differences and the range of children's abilities. The present study hypothesized that perceived stress would mediate the associations of job demands and resources (both job-

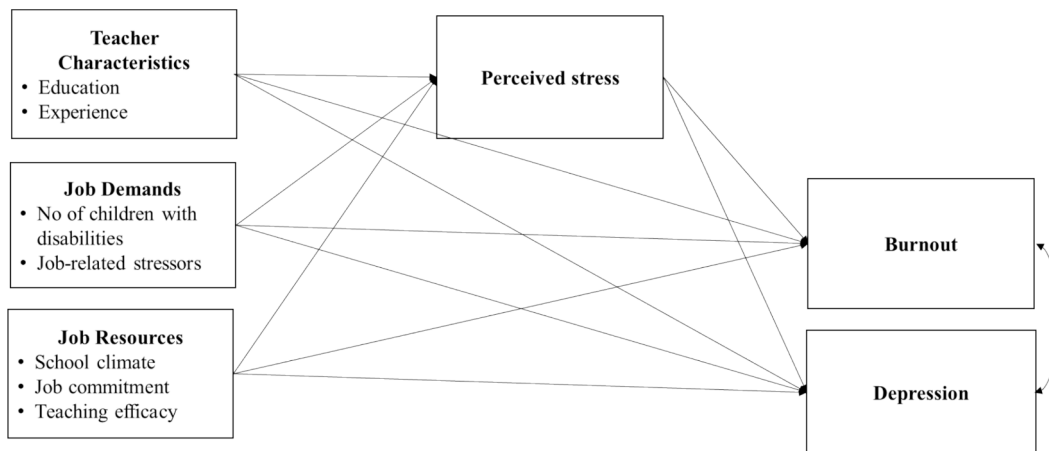


Fig. 1. Study Conceptual Model.

related and internal resources) with job burnout and depression. Previous studies have found that teachers' perceived stress is associated with their job burnout (Jennings & Greenberg, 2009) and depressive symptoms (Whitaker et al., 2013) and mediates the links between teachers' internal resources and their job burnout (Schwarzer & Hallum (2008)). Thus, we examined this mediational model with our total sample and separately for the ECE and ECSE samples. Earlier studies also have found that teachers' job burnout and depressive symptoms are positively correlated with each other (Jeon et al., 2018). In our model, the covariance between teachers' job burnout and depressive symptoms is estimated, and we also included teachers' level of education and teaching experience as covariates. Our specific study hypotheses are as follows:

- ECE teachers' perceived stress, job burnout, and depressive symptoms are different from those of ECSE teachers.
- ECE and ECSE teachers' job demands, job resources, and internal resources are associated with their perceived stress. These relationships differ between ECE and ECSE teachers.
- ECE and ECSE teachers' job demands, job resources, internal resources, and perceived stress are associated with their job burnout. These relationships differ between ECE and ECSE teachers.
- ECE and ECSE teachers' job demands, job resources, internal resources, and perceived stress are associated with their depressive symptoms. These relationships differ between ECE and ECSE teachers.
- ECE and ECSE teachers' perceived stress mediates the associations of their job demands, job resources, and internal resources to their job burnout and depressive symptoms. These mediation links differ between ECE and ECSE teachers.

2. Method

2.1. Participants

Participants in this study included 143 ECE teachers and 121 ECSE teachers in a western state in the United States. ECE teachers were from state-funded ($n = 31$, 21.7 %), Early Head Start ($n = 24$, 16.8 %), Head Start ($n = 6$, 4.2 %), non-profit ($n = 13$, 9.1 %), faith-based ($n = 15$, 10.5 %), and private programs ($n = 54$, 37.8 %). ECSE teachers were from inclusive classrooms ($n = 41$, 33.9 %), self-contained ECSE classrooms ($n = 40$, 33.1 %), and ECSE classrooms for children with autism ($n = 40$, 33.1 %) in a large school district. In order to recruit representative ECE and ECSE teachers in the state, we obtained a list of state-registered ECE programs in a large metropolitan area and an ECSE teachers' email list from a large school district.

Within the ECE sample, 44 programs (program response rate = 46.8 %) agreed to participate in our study, and we distributed research study information to their teachers. Of those teachers, 284 indicated their willingness to participate in the study by providing us with their email information so they could be sent the survey. Of those, after screening participants for eligibility requirements (being a head teacher), a total of 143 ECE teachers completed the questionnaires (response rate = 50.3 %).

Similarly, for the ECSE sample, the researchers used an inclusive list of 372 ECSE teachers working with children with disabilities in the school district who were between the ages of 3 and 5 years old. One hundred classroom teachers from each classroom setting type were then invited to participate so as to ensure that there were representative samples of each of the three classroom types. A final sample of 121 ECSE teachers comprised the population of the ECSE teacher group with a response rate of 40 %. Prior to conducting the online survey, we obtained approval from the University of Nevada, Reno Institutional Review Board (Protocol #1029523-16) on September 13, 2018, and from the Research Review Committee of the Clark County School District on October 3, 2018. Participation in the study was voluntary, and informed consent was electronically obtained from all teacher participants. Each participant received an e-gift card upon completing the survey.

Both ECE and ECSE teachers' demographic characteristics are presented in Table 1. The vast majority of the ECE teacher participants were female ($n = 140$, 98 %), European Americans (74 %). Of the ECE teachers, 53 % held a high school diploma as their highest degree, while 35 % had a bachelor's degree or higher, and 11 % had an associate degree. Among the ECE teachers, 43 % had worked in the field for less than 5 years, 32 % had worked in the field for 5–10 years, and 25 % had worked in the field for more than 10 years. Of the surveyed teachers, 38 % had experience working with children with disabilities in the classroom.

Of all the ECSE teachers who completed the survey, 95.9 % were female and 76 % were European Americans. The highest education level completed by participating teachers was reported as 24.8 % holding a bachelor's degree, 68.6 % with a master's degree, 4.1 % were education specialists, and 2.5 % had a doctoral degree. A little more than a third of the teachers (34 %) majored in special education, and 32 % majored in ECSE. Almost 22 % of the ECSE teachers had worked in the field for less than 5 years, whereas about 45 % of ECSE teachers had worked in their field for more than 10 years.

2.2. Measures

The online questionnaire included job-related stress (Curbow et al., 2000), school/program climate (Battistich et al., 1997), job commitment

Table 1

Descriptive Statistics of Participants' Characteristics and for Study Variables by ECE and ECSE Teachers.

Variable	ECE				ECSE			
	n/M	%/SD	Min	Max	n/M	%/SD	Min	Max
Age (years)	32.77	13.28	17	68	45.12	11.80	21	72
Gender								
Female	140	97.9			116	95.9		
Male	2	1.4			5	4.1		
Other	1	0.7						
Ethnicity								
Caucasian	106	74.41			92	76.0		
Latino	18	12.6			11	9.1		
African American/Black	3	2.1			7	5.8		
Asian American	4	2.8			6	5.0		
American Indian/Alaskan	3	2.1						
Native/Pacific Islander	2	1.4			1	0.8		
Biracial/Multiracial	7	4.9			4	3.3		
Education								
Less than HS/GED	1	0.7						
High School/GED	33	23.1						
Some college	43	30.1						
Associates degree	16	11.2						
Bachelor's degree	37	25.9			30	24.8		
Master's degree	13	9.1			83	68.6		
Education specialist					5	4.1		
Doctoral degree					3	2.5		
Major								
Early childhood education	69	48.3			14	11.6		
Early childhood special education	3	2.1			39	32.2		
Special education	2	1.4			41	33.9		
N/A	32	22.4						
Other	37	25.9						
Experience in the field								
< 5 years	61	42.7			26	21.5		
5 – 10 years	46	32.2			41	33.9		
>10 years	36	25.2			54	44.6		
No. of children with special needs ^a	0.97	1.56	0	5+	8.28	2.39	2	10
Job-related stressors	3.22	0.76	1.33	5	3.24	0.66	1	4.83
School climate	3.72	0.88	1.38	5	3.75	0.79	1.69	4.92
Job commitment	36.23	6.94	17	50	37.45	8.14	14	50
Teaching efficacy	4.60	0.44	2.89	5	4.54	0.47	2.77	5
Perceived stress	24.95	6.97	10	54	23.51	5.75	10	40
Job burnout	1.49	0.77	0	3.45	1.55	0.76	0.05	3.18
Depressive symptoms	8.78	7.85	0	43	8.25	7.72	0	37

Note. ECE = early childhood education teachers ($n = 143$); ECSE = early childhood special education teachers ($n = 121$). ^a Total number of children with disabilities in the classroom.

(Blau, 1985; Hackett et al., 2001; Landy & Guion, 1970), teaching efficacy (Early Childhood Teaching Inventory: Van de Wiele, 2001), perceived stress (Cohen et al., 1993), job burnout (Maslach & Jackson, 1981), and depression inventory (Beck, et al., 1961), as well as teachers' demographic and classroom characteristics.

2.3. Job demands

Teachers' job demands were captured using the Child Care Worker Job Stress Inventory (CCW-JSI; Curbow et al., 2000). This measure was designed to assess the job-related stress of child care workers and providers. The CCW-JSI utilizes a 5-point Likert scale asking teachers to indicate how often something happens in their work. The measure includes such statements as "I feel like I have to be a parent and a teacher to the children," and "All of the children need attention at the same time." We used the Job Demands subscale (six items) of an abbreviated version of the CCW-JSI, which is consistent with what earlier studies had done (Bryant et al., 2009; Jeon et al., 2019). This subscale illustrates teachers'

job-related stress related to working with both children and their families in the classroom setting. To ensure each item within the measure was appropriate for a given teacher population, we modified an item slightly to better represent the terms used in ECSE settings. Specifically, in the ECSE questionnaire we replaced a term found in the ECE questionnaire: "on child care" with "in the classroom." The ECE version used the original terms. Curbow et al. (2000) examined the reliability as well as convergent and discriminant validity of the CCW-JSI. Researchers found that ECE teachers' job-related stress scores were associated with depressive symptoms and professional motivation (Jeon et al., 2019) as well as with preschoolers' expulsions (Gilliam & Shahar, 2006). According to Bryant et al. (2009), internal consistency was 0.84 with Job Demands in a sample of ECE teachers. Cronbach's α s were 0.78 and 0.68 for the current ECE and ECSE teacher samples, respectively. Teachers' job demands scores were calculated by averaging all six items of the Job Demands subscale.

2.4. School/Program climate

The Teachers' Sense of the School (Battistich et al., 1997) measure was used to assess teachers' perceptions of their school/program environment. Comprising 13 items, this scale was developed to evaluate several indicators of a positive school climate including teacher cooperation and sharing of responsibilities, supportive relationships between teachers and administrators, and their level of authority within school-wide decision-making processes. The scale indicates teachers' level of agreement with a statement on a 5-point Likert scale from 1 (*strongly disagree*) to 5 (*strongly agree*). Items include, "This school seems like a big family; everyone is so close and cordial," and "Teachers take a major role in shaping the school's norms, values, and practices." According to Battistich et al. (1997), this measure has good internal consistency ($\alpha = 0.89$). In addition, Guo et al. (2011) reported that preschool teachers' sense of school community was related to children's literacy gains. In the current ECE and ECSE teacher samples, Cronbach's α s were 0.92. The mean scores of the Teachers' Sense of the School were calculated. Higher scores mean that teachers perceive their school/program environment as being more positive and supportive.

2.5. Job commitment

The current study used a modified measure of Hackett et al.'s (2001) occupational commitment scale. The original measure was adapted from two different scales, using Blau's (1985) career commitment measure and Landy's and Guion's (1970) work motivation measure. Our measure includes 10 items in which teachers rated their level of agreement with each statement on a 5-point Likert scale from 1 (*strongly disagree*) to 5 (*strongly agree*). Example items include, "If I could get a different job from being an ECE/ECSE teacher that paid the same amount, I would take it," and "I like this profession too much to give it up." According to Blau (1985), occupational commitment was negatively related to an intention to quit, and Hackett et al. (2001) reported that individuals' involvement in their work predicted their job commitment. In the current ECE and ECSE teacher samples, Cronbach's α s were 0.74 and 0.70, respectively. The mean scores of the 10 items were calculated with higher scores indicating a stronger job commitment.

2.6. Teaching-Efficacy

Teachers' perceived teaching efficacy was measured by using the Early Childhood Teaching Inventory (Van de Wiele, 2001). This 19-item scale asked teachers to rate their ability to do general tasks on a 5-point scale from 1 (*I cannot do this*) to 5 (*I'm sure I can do this*). These statements included items related to working with students ("Manage behavior problems among the children in my classroom"), working with families ("Maintain a working relationship with a parent even if I disagree with his/her attitudes about child care and education"), and making use of professional resources ("Find resource materials to help me when I need information to plan a learning activity"). To ensure each item within the measure was appropriate for the ECSE teacher population, we slightly modified items so as to better represent the terms used in ECSE settings. Specifically, within ECSE we replaced two terms, "child care" with "school" and "child care teacher" with "ECSE teacher." Hughes-Belding et al. (2012) found that the Early Childhood Teaching Inventory had good internal consistency (0.92), and providers' professional motivation was related to home-based care quality. In the current sample, Cronbach's α s were 0.93 and 0.95 for ECE and ECSE teachers, respectively. Teachers' teaching efficacy scores were calculated by averaging all items.

2.7. Perceived stress

The Perceived Stress Scale (PSS; Cohen et al., 1983) provides an overarching assessment of global stress that can be utilized across

diverse individuals and groups (Jeon et al., 2018; Lee & Jeong, 2019; Wagner et al., 2013). The PSS includes 14 items asking how often an individual had certain thoughts or feelings during the last month on a 5-point scale from 0 (*never*) to 4 (*very often*). Examples from the PSS include, "In the last month, how often have you been upset because of something that happened unexpectedly?" and "In the last month how often have you dealt successfully with irritating life hassles?" In the current study, this scale was used to appraise the degree to which ECE and ECSE teachers' lives were stressful, unpredictable, or overwhelming. The internal consistency of the PSS was reported as 0.85 (Cohen et al., 1983). Cronbach's α s for the current ECE and ECSE teacher samples were 0.80 and 0.88, respectively. After reverse coding seven negative items, summative scores for all of these items were calculated.

2.8. Job burnout

To evaluate teachers' feelings related to their work, the current study used the Maslach Burnout Inventory for Educators (Maslach & Jackson, 1981), which asked teachers to rate how often they experienced the feelings in each statement on a scale from 0 (*never*) to 6 (*every day*). The 22-item scale includes three subscales: exhaustion (9 items), depersonalization (5 items), and personal accomplishment (8 items). The emotional exhaustion subscale assesses feelings of emotional fatigue and being overextended in their work, while the depersonalization subscale assesses the teachers' impersonal responses towards their students (e.g., indifferent, cold, or negative attitudes) and their tendency to withdraw emotionally (Maslach & Leiter, 2016). The final subscale, personal accomplishment, measures teachers' feelings of success and fulfillment in their teaching. According to Iwanicki and Schwab (1981), internal consistencies for emotional exhaustion ($\alpha = 0.90$), depersonalization ($\alpha = 0.79$), and personal accomplishment ($\alpha = 0.76$) were acceptable, and these are consistent with the findings of other professionals in the human service field. Intercorrelations across subscales were between 0.75 and 0.94 with a mean of 0.87 (Iwanicki & Schwab, 1981). After the items of the personal accomplishment subscale were reversed coded, we combined these three subscales by averaging the item scores to measure both ECSE and ECE teachers' overall job burnout. Cronbach's α for the current ECE and ECSE teacher samples were 0.86 and 0.87, respectively.

2.9. Depressive symptoms

Beck's Depression Inventory (BDI; Beck et al., 1961) was used to capture teachers' cognitive, affective, and somatic depressive symptoms and attitudes. This inventory asks individuals to respond to 21 items with the degree of severity on a fully anchored 4-point scale from 0 to 3. Examples of the items in the inventory include the loss of energy and work inhibition "0 (*I can work about as well as before*) to 3 (*I can't do any work at all*)" and mood, "0 (*I do not feel sad*) to 3 (*I am so sad and unhappy I can't stand it*)." A summative score of all these items in the inventory indicates the overall level of depressive symptoms. According to the Center for Cognitive Therapy (as cited in Beck et al., 1997), cut-off scores for normal or minimal depression is < 14, mild depression is 14–19, moderate depression is 20–28, and severe depression is > 28. The average of BDI summative scores for the current samples were normal or minimal depression (77.6 % of ECE teachers and 80.2 % of ECSE teachers). The average internal consistency was 0.87 across the different sample populations, and it was correlated with the Hamilton Psychiatric Rating Scale and Depression (Beck et al., 1997). Cronbach's α s for the current ECE and ECSE teacher samples were 0.88 and 0.91, respectively.

2.10. Data analysis plan

First, we examined t-tests to compare the means of the study variables between ECE and ECSE teachers. Second, we looked at bivariate correlations between study variables using IBM SPSS Statistics version

28. Finally, we used path modeling to examine relations among teachers' work-related demands and resources, perceived stress, job burnout, and depressive symptoms using *Mplus* version 8 (Muthén & Muthén, 1998–2018). Our path model included the mediation paths of teachers' perceived stress in relation to their job demands and resources with their job burnout and depressive symptoms. We also compared pathways of ECE and ECSE teachers' perceived stress, job burnout, and depressive symptoms.

3. Results

3.1. Preliminary analyses

Descriptive statistics for the study variables are presented in Table 1. There was no difference in the means of the study variables (perceived stress, job burnout, and depressive symptoms) between ECE and ECSE teachers, despite different levels of education, amounts of experience working with children, and numbers of children with disabilities in the classroom. ECSE teachers ($M = 5.78$, $SD = 0.47$) were more likely to have a higher level of education than ECE teachers, ($M = 3.66$, $SD = 1.34$), $t(182) = 17.62$, $p < 0.001$, Cohen's $d = 2.04$. ECSE teachers also tended to have more experience working with children than ECE teachers, $\chi^2(2) = 16.25$, $p < 0.001$. ECSE teachers ($M = 3.88$, $SD = 1.74$) tended to work with more children with disabilities than ECE teachers, ($M = 0.97$, $SD = 1.56$), $t(262) = 14.36$, $p < 0.001$, Cohen's $d = 1.77$.

Table 2 presents bivariate correlations between the study variables by ECE and ECSE teachers. ECE teachers' educational attainment and working experience were negatively related to their perceived stress, job burnout, and depressive symptoms, and the number of children with disabilities in their classroom was positively related to their job burnout and depressive symptoms. However, ECSE teachers' educational attainment or the number of children with disabilities were not related to their perceived stress, job burnout, or depressive symptoms. Both ECE and ECSE teachers' job-related stressors, school climate, and teaching efficacy had similar relationships with their perceived stress, job burnout, and depressive symptoms. One exception we found was that job commitment was negatively associated with depressive symptoms, but only for ECSE teachers.

3.2. Path analyses

To test whether the path coefficients of our study path model (Fig. 1) differ between ECE and ECSE teachers, we compared a fully constrained model, in which all paths were set to be equal between the two groups, to the fully saturated model where all paths were estimated for both the groups. The goodness-fit-index for the fully constrained model was $\chi^2(37) = 95.16$, $p < 0.001$, indicating that the paths in the study model significantly vary between ECE and ECSE teachers. We reported the path coefficients from the fully saturated model.

Table 2
Bivariate Correlation Among Study Variables.

	1	2	3	4	5	6	7	8	9	10
1. Education	—	0.41**	0.23*	−0.07	−0.01	−0.15	0.11	−0.08	−0.05	−0.18
2. Experience	0.28**	—	0.14	−0.06	0.12	0.05	0.26**	−0.27**	−0.08	−0.18*
3. No of children ^a	0.21*	0.09	—	0.14	−0.05	−0.15	0.02	0.07	0.17	0.02
4. Job-related stressors	−0.20*	−0.14	0.12	—	−0.16	−0.18*	−0.08	0.31**	0.35**	0.28**
5. School climate	0.14	0.15	−0.15	−0.46**	—	0.32**	0.30**	−0.35**	−0.39**	−0.28**
6. Job commitment	0.04	0.23**	0.03	−0.43**	0.37**	—	0.28**	−0.28**	−0.53**	−0.30**
7. Teaching efficacy	0.15	0.10	0.01	−0.17*	0.26**	0.22**	—	−0.30**	−0.34**	−0.30**
8. Perceived stress	−0.25**	−0.20*	0.13	0.33**	−0.32**	−0.18*	−0.33**	—	0.57**	0.68**
9. Job burnout	−0.17*	−0.18*	0.30**	0.57**	−0.41**	−0.46**	−0.37**	0.59**	—	0.60**
10. Depression	−0.23**	−0.16	0.24**	0.26**	−0.21*	−0.11	−0.28**	0.70**	0.61**	—

Note. The results for early childhood special education teachers ($n = 121$) are displayed above the diagonal. The results for early childhood education teachers ($n = 143$) are displayed below the diagonal. ^a Total number of children with disabilities in the classroom.

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

Table 3 displays path coefficients of job demands and resources for the combined group of teachers, and then separately for ECE and ECSE teachers. Based on path coefficients for the combined ECE and ECSE teachers, teachers who had higher educational levels and more years of experience perceived they experienced less stress ($B = -0.88$, $SE = 0.30$), but their backgrounds were not associated with their job burnout or depressive symptoms. Those who worked in classrooms with more children with disabilities were more likely to experience job burnout ($B = 0.07$, $SE = 0.02$) and depressive symptoms ($B = 0.42$, $SE = 0.20$), but the number of children with disabilities was not related to their perceived stress. Teachers who experienced more job-related stressors were more likely to perceive stress ($B = 1.62$, $SE = 0.54$) and experience more job burnout ($B = 0.22$, $SE = 0.05$). Teachers who reported having a more positive school climate perceived that they had less stress ($B = -1.27$, $SE = 0.46$), and those who had higher scores for job commitment were less likely to have experienced job burnout ($B = -0.03$, $SE = 0.01$). Those who had higher teaching efficacy were less likely to perceive stress ($B = -2.84$, $SE = 0.81$) and experienced less job burnout ($B = -0.21$, $SE = 0.07$). Teachers' perceived stress was positively related to their job burnout ($B = 0.04$, $SE = 0.01$) and depressive symptoms ($B = 0.77$, $SE = 0.06$).

We simultaneously examined the same mediation model between ECE and ECSE teachers separately using the group variable. We found some similarities and differences in the pathways of job burnout and depressive symptoms between ECE and ECSE teachers (Table 3). Among ECE teachers (see Fig. 2), their highest educational level was negatively related to their perceived stress, and the number of children with disabilities was positively related to their job burnout and depressive symptoms. Their job-related stressors were positively related to their perceived stress and job burnout. Their job commitment was negatively related to their job burnout, and their teaching efficacy was negatively related to their perceived stress and job burnout. However, their school/program climate was not related to their perceived stress, job burnout, or depressive symptoms.

Among ECSE teachers (see Fig. 3), their highest achieved educational level was negatively associated only with having depressive symptoms, whereas their years of experience were negatively associated with only their perceived stress. ECSE teachers who reported more job-related stressors were more likely to perceive stress, whereas those who perceived that they had a more supportive school climate were less likely to perceive stress. Those who had higher job commitment were less likely to experience job burnout. Their teaching efficacy was not related to their perceived stress, job burnout, or depressive symptoms. None of the ECSE teachers' job demands or job-related resources were related to their depressive symptoms.

3.3. Mediation analysis

Mediation analysis results (see Table 4) show that ECE teachers'

Table 3

Path Coefficients of Job Demands and Resources on Perceived Stress, Job Burnout, and Depression.

Predictor	Perceived stress (a)			Burnout (c1')			Depression (c2')		
	<i>B</i>	<i>B SE</i>	β	Predictor	<i>B</i>	<i>B SE</i>	β	Predictor	<i>B</i>
Total									
Teacher Characteristics									
Education	-0.88**	0.30	-0.20	-0.03	0.03	-0.07	-0.43	0.30	-0.08
Experience	-0.96*	0.46	-0.12	0.03	0.04	0.04	-0.01	0.46	0.00
Job Demands									
No. children ^a	0.34+	0.20	0.11	0.07***	0.02	0.21	0.42*	0.20	0.12
Job-related stressors	1.62**	0.54	0.18	0.22***	0.05	0.21	0.31	0.54	0.03
Job Resources									
School climate	-1.27**	0.46	-0.16	-0.05	0.04	-0.06	0.32	0.47	0.04
Job commitment	-0.03	0.05	-0.04	-0.03***	0.01	-0.29	-0.04	0.05	-0.04
Teaching efficacy	-2.84***	0.81	-0.20	-0.21**	0.07	-0.13	-1.32	0.82	-0.08
Mediator									
Perceived stress (b)				0.04***	0.01	0.38	0.77***	0.06	0.64
ECE									
Teacher Characteristics									
Education	-0.87*	0.41	-0.17	-0.02	0.03	-0.04	-0.53	0.38	-0.09
Experience	-0.87	0.67	-0.10	-0.01	0.05	-0.01	-0.28	0.60	-0.03
Job Demands									
No. of children ^a	0.60+	0.34	0.13	0.12***	0.03	0.24	0.93**	0.31	0.18
Job-related stressors	1.59*	0.81	0.17	0.29***	0.06	0.29	0.27	0.73	0.03
Job Resources									
School climate	-1.02	0.68	-0.13	0.01	0.05	0.01	0.61	0.61	0.07
Job commitment	0.02	0.09	0.02	-0.03***	0.01	-0.25	0.01	0.08	0.01
Teaching efficacy	-3.81**	1.20	-0.24	-0.26**	0.10	-0.15	-1.15	1.11	-0.07
Mediator									
Perceived stress (b)				0.04***	0.01	0.36	0.72***	0.08	0.64
ECSE									
Teacher Characteristics									
Education	-0.08	1.07	-0.01	-0.19+	0.11	-0.12	-2.59*	1.19	-0.16
Experience	-1.40*	0.65	-0.19	0.12+	0.07	0.13	0.79	0.73	0.08
Job Demands									
No. of children ^a	0.15	0.27	0.04	0.04	0.03	0.09	-0.11	0.30	-0.03
Job-related stressors	1.97**	0.70	0.23	0.15+	0.08	0.13	0.64	0.80	0.06
Job Resources									
School climate	-1.56*	0.62	-0.21	-0.10	0.07	-0.11	0.00	0.70	0.00
Job commitment	-0.08	0.06	-0.12	-0.03***	0.01	-0.34	-0.12+	0.07	-0.13
Teaching efficacy	-1.67	1.06	-0.14	-0.18	0.11	-0.11	-1.29	1.18	-0.08
Mediator									
Perceived stress (b)				0.05***	0.01	0.38	0.82***	0.10	0.61

Note. ^a Total number of children with disabilities in the classroom. ECE = early childhood education teachers; ECSE = early childhood special education teachers.
+ $p < 0.10$. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

teaching efficacy was negatively related to their job burnout ($B = -0.15$, $SE = 0.05$) and depressive symptoms ($B = -2.76$, $SE = 0.91$) through their perceived stress. However, ECE teachers' perceived stress did not mediate the associations between job demands (i.e., the number of children with disabilities and job-related stressors) or resources (i.e., school climate and job commitment) and teachers' depressive symptoms or job burnout. Among ECSE teachers, their job-related stressors were positively related to their job burnout ($B = 0.10$, $SE = 0.04$) and depressive symptoms ($B = 1.62$, $SE = 0.61$) through their perceived stress, whereas their school climate was negatively related to their job burnout ($B = -0.08$, $SE = 0.04$) and depressive symptoms ($B = -1.28$, SE

$= 0.05$) through their perceived stress. The number of children with disabilities in their classroom, ECSE teachers' job commitment, or teaching efficacy was not related to their job burnout or depressive symptoms through their perceived stress.

4. Discussion

The current study compares and examines ECE and ECSE teachers' perceived stress, job burnout, and depressive symptoms using a mediational model with group comparisons. Given the limited research that has been done that includes both ECE and ECSE teachers' well-being,

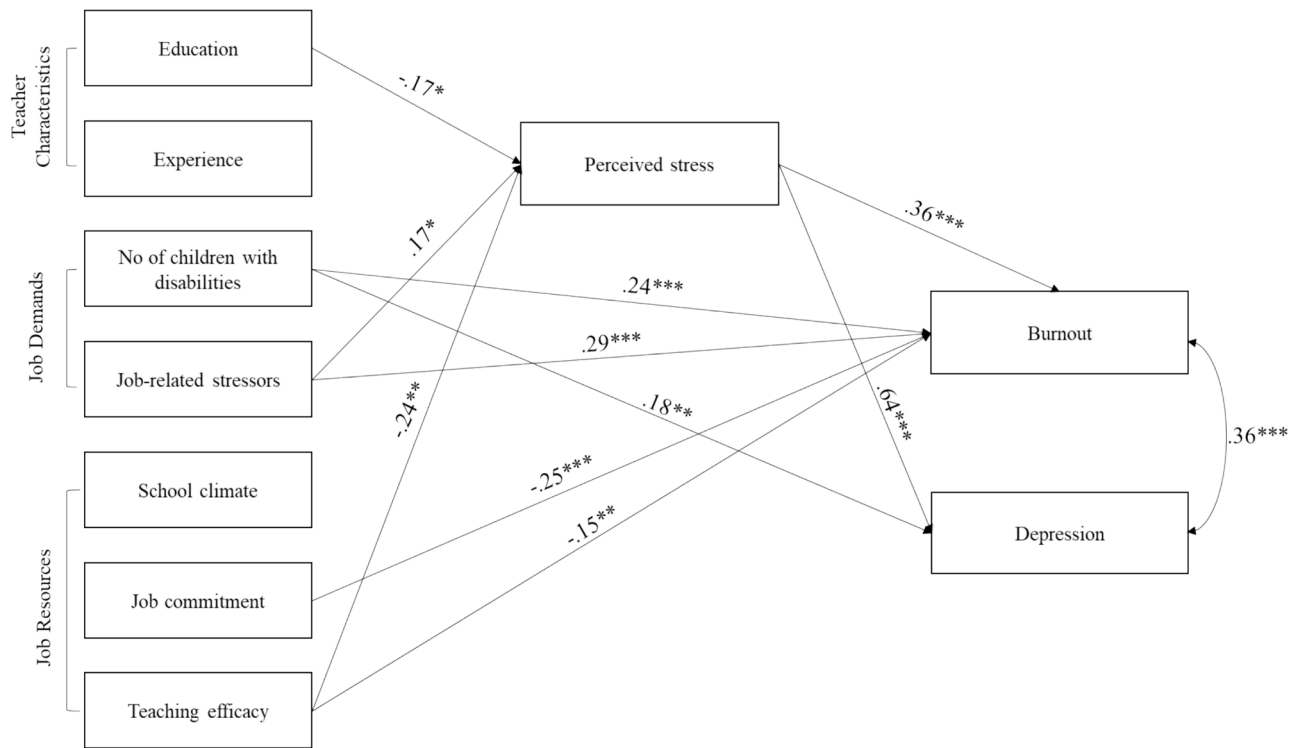


Fig. 2. Path Coefficients Predicting Perceived Stress, Job Burnout, and Depressive Symptoms for Early Childhood Education Teachers. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

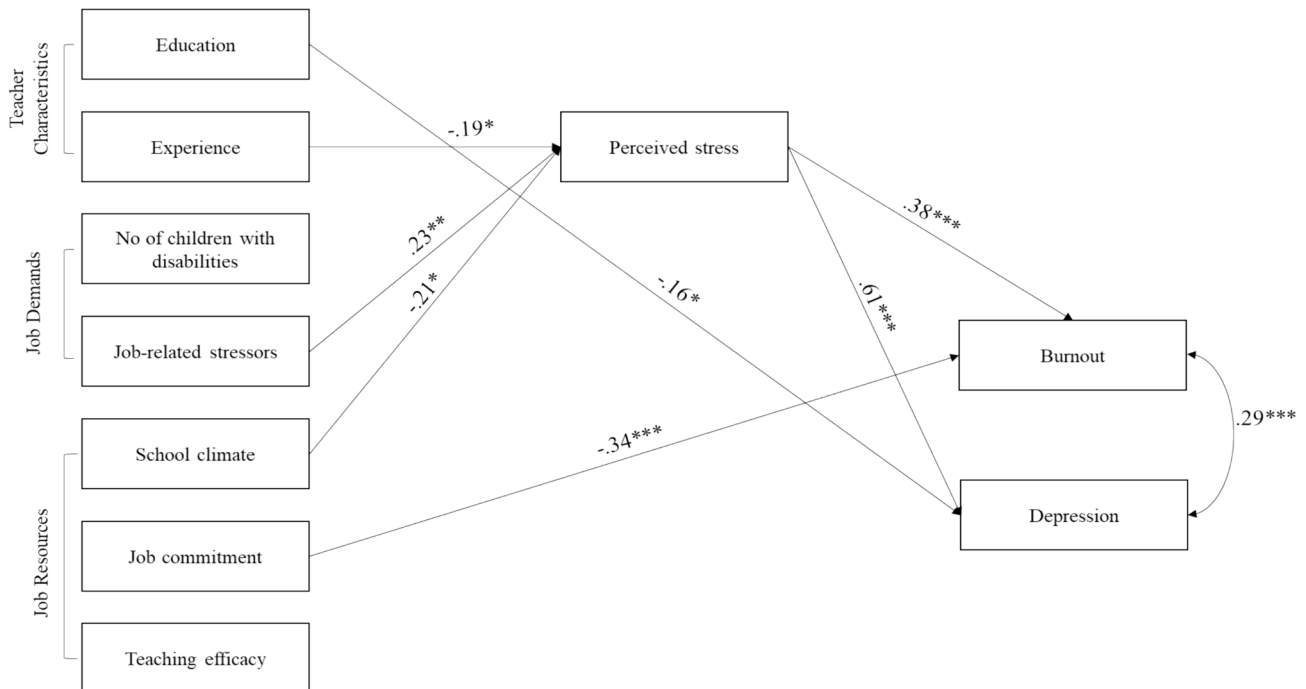


Fig. 3. Path Coefficients Predicting Perceived Stress, Job Burnout, and Depressive Symptoms for Early Childhood Special Education Teachers. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

our findings offer a unique contribution to the literature on job demands, job resources, and the well-being of teachers who work with young children. We found that there was no difference in the teachers' perceptions of work-related stressors, perceived stress, job burnout, and depressive symptoms between ECE and ECSE teachers, even though their educational levels, amount of experience working in their field,

and the number of children with disabilities in their classrooms differed. Our findings are inconsistent with those of previous studies showing that Canadian teachers who worked with children with autism perceived less stress (Cappe et al., 2017), and Israeli special education teachers experienced less job burnout (Hazan-Liran & Karni-Vizer, 2023) than did general education teachers including preschool teachers. These

Table 4

Indirect Effects of Job Demands and Resources on Burnout and Depressive Symptoms Through Perceived Stress by ECE and ECSE Teachers.

Predictors	ECE						ECSE					
	Burnout			Depression			Burnout			Depression		
	<i>B</i>	<i>B SE</i>	β	<i>B</i>	<i>B SE</i>	β	<i>B</i>	<i>B SE</i>	β	<i>B</i>	<i>B SE</i>	β
Job Demands												
No. of children ^a	0.02+	0.01	0.05	0.43+	0.25	0.09	0.01	0.01	0.02	0.12	0.22	0.03
Job-related stressors	0.06+	0.03	0.06	1.15+	0.60	0.11	0.10*	0.04	0.09	1.62**	0.61	0.14
Job Resources												
School climate	-0.04+	0.03	-0.05	-0.74	0.50	-0.08	-0.08*	0.04	-0.08	-1.28*	0.05	-0.13
Job commitment	<0.01	<0.01	0.01	0.02	0.06	0.01	>0.01	>0.01	-0.05	-0.07	0.05	-0.07
Teaching efficacy	-0.15**	0.05	-0.09	-2.76**	0.91	-0.16	-0.08	0.06	-0.05	-1.37	0.89	-0.08

Note. ECE = early childhood education teachers; ECSE = early childhood special education. ^a Total number of children with disabilities in the classroom.+ $p < 0.10$. * $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$.

researchers have explained their findings with the difference in teachers' motivation (Cappe et al., 2017) and training (Hazan-Liran & Karni-Vizer, 2023), but we did not find that difference in internal resources between our ECE and ECSE teachers even though their levels of education differed. Further investigation should test whether ECE and ECSE teachers with different educational backgrounds and years of work experience in their fields, experience similar levels of job demands (e.g., moderate levels of job-related stressors) and resources and psychological and professional well-being.

There were some differences in associations among job demands and resources and psychological and professional well-being between ECE and ECSE teachers. We found that ECE teachers' teaching efficacy was negatively related to their perceived stress and job burnout, and the links between their teaching efficacy and job burnout and depression were mediated by that perceived stress. For ECE teachers, the number of children in their classroom with disabilities and job-related stressors were positively related to their job burnout, whereas their job commitment was negatively associated with their job burnout. For them, school climate was not a significant factor for the teacher well-being variables. For ECSE teachers, however, teaching efficacy was not associated with their perceived stress nor with job burnout. A supportive and collaborative school climate was negatively associated only with ECSE teachers' perceived stress, but it did not directly predict their job burnout or their depressive symptoms. ECSE teachers' perceived stress mediated the links of their job-related stressors with job burnout and depressive symptoms along with the links of their school climate with their job burnout and depressive symptoms. Overall, consistent with the JD-R model (Bakker & Demerouti, 2017), we found that various job demands and resources predicted the psychological well-being of both ECE and ECSE teachers, despite the similarities and differences between ECE and ECSE teachers.

First, we found that both ECE and ECSE teachers who reported having more job-related stressors were more likely to have a higher level of perceived stress. These findings confirm previous findings which mostly focused on ECE teachers' job-related demands and stressors related to their perceived stress (Jeon et al., 2018; Tsai et al., 2006) and add evidence about ECSE teachers. In our study, there were some differences in the associations between job-related stressors and teachers' well-being between ECE and ECSE teachers. Among ECE teachers, job-related stressors were directly related to their job burnout, which is consistent with the findings in other studies of ECE teachers (Jeon et al., 2018) along with those of general education teachers in Norway (Skaalvik & Skaalvik, 2011) and in Cyprus (Kokkinos, 2007). Further, in our study, ECSE teachers' job-related stressors were indirectly linked to their job burnout and their depressive symptoms through their perceived stress.

Previous studies (e.g., Jeon et al., 2018; Skaalvik & Skaalvik, 2011; Tsai et al., 2006) did not explore job-related stressors' mediational relationships with teachers' well-being. Job-related stressors are part of an

individual's stressors, directly linked to their perceived stress, which, in turn, are related to job burnout and their depressive symptoms. However, it is perplexing that our study found these mediational links among ECSE teachers but not among ECE teachers. To better understand the relationships between ECE and ECSE teachers' job-related stressors and professional and psychological well-being, further longitudinal and experimental investigation is necessary to examine which job-related demands and stressors are related to different aspects of their well-being, and why and how they have disparate relationships with various aspects of well-being (e.g., direct links or mediational links).

Second, ECE teachers who had more children with disabilities in their classrooms tended to experience more job burnout and depressive symptoms, whereas the number of children with disabilities in their classroom was not related to ECSE teachers' job burnout and depressive symptoms. This finding is similar to that of Schaack et al.'s study (2020), which found that ECE teachers who reported having more children with behavior difficulties in their classrooms experienced more job burnout. Other researchers have found that the levels of classroom problem behavior were directly related to ECE teachers' perceived stress and job burnout for ECE teachers (Kwon et al., 2021; Jeon et al., 2018) as well as to Norwegian elementary and middle school teachers' job burnout (Skaalvik & Skaalvik, 2011).

Even though there are a few studies on the links between the number of children with disabilities in a classroom and teachers' well-being, we can suggest some reasons why the current study found a difference in those links between ECE and ECSE teachers. Compared to ECSE teachers, ECE teachers generally work with more children in their classrooms (Sacco, 2011) and may experience more job burnout and depressive symptoms when they have children with disabilities. ECE teachers tend to have limited specialized training on how to work with children with disabilities and how to meet those children's different developmental needs (Cook, 2002; Shippen et al., 2005; Swain et al., 2012). ECE teachers' lack of training and preparation for dealing with children with disabilities may lead to their experiencing more job burnout and to having more depressive symptoms.

On the other hand, we did not find a link between the number of children with disabilities in teachers' classrooms and ECE teachers' general perceived stress. Rather, having more children with disabilities in their classroom was directly related to their job burnout and depressive symptoms. It is not clear what the reasons are for this lack of a direct association between the number of children with disabilities in an ECE teacher's classroom and their perceived stress. However, we speculate that perceived stress, which may be more momentarily accommodated (McGonigal, 2015), may be associated with global stressors, whereas job burnout is tied mostly to the work environment (Iacovides et al., 2003). Furthermore, in our study, ECE teachers who had higher levels of education were more likely to work with children with disabilities in their classroom and were less likely to perceive stress. This indicates that other extraneous factors may be associated with ECE

teachers' having children with disabilities in their classroom and their perceived stress. Our findings suggest that additional investigation of the relationship between the number of children with disabilities in teacher's classrooms and teachers' well-being is warranted.

Third, we found that ECSE teachers who perceived their school as being supportive and a collaborative workplace tended to perceive that they had less stress which, in turn, led them to experience less job burnout and fewer depressive symptoms (mediational links). However, we did not find the same relationship among ECE teachers. Our findings indicate that a supportive and collaborative school climate is an important job resource that can only promote the well-being of ECSE teachers. Our findings about ECSE teachers are consistent with earlier findings which showed that a supportive and collaborative school climate is related to ECE teachers having less job burnout and fewer depressive symptoms (Jeon et al., 2018; Schaack et al., 2020) as is also the case with elementary and middle school teachers' (Skaalvik & Skaalvik, 2021). Unlike the findings of other studies including ECE teachers (e.g., Jeon et al., 2018; Schaack et al., 2020), we did not find the same relationships between school climate and the well-being of ECE teachers. We can speculate about explanations for these differences in the association of school climate with psychological well-being between ECE and ECSE teachers. ECSE teachers need to work more often with other teachers, staff members (e.g., occupational therapists, physical therapists, language pathologists), and administrators to meet the specific needs and accommodations for children with disabilities than ECE teachers (e.g., Adera & Bullock, 2010). Hence a positive and collaborative school climate plays an essential role in the psychological and professional well-being of ECSE teachers but not for ECE teachers. We also suspect that ECE teachers are more likely to work in smaller programs with fewer classrooms compared to those of ECSE teachers who often work in larger school settings, which means that ECE teachers may not be as affected by a supportive program climate as ECSE teachers. Additionally, ECE teachers in private programs may more easily leave their position for other positions in different programs or find another occupation other than being an ECE teacher, as opposed to ECSE teachers who work for a public school. We also speculate that the variation between our findings and those of Jeon et al. (2018) may be due to differences among ECE teachers sampled. Compared to Jeon et al., ECE teachers in our study were from diverse early childhood programs, including many private infant and toddler programs, Early Head Start, and faith-based programs, whereas those in Jeon et al. were from public preschool and pre-kindergarten classrooms. The school climate in private and smaller infant and toddler programs may not be as related to ECE teachers' well-being as compared to that of teachers in formal preschool or public pre-kindergarten programs.

In contrast to previous studies (e.g., Jeon et al., 2018; Schaack et al., 2020; Skaalvik & Skaalvik, 2021), we simultaneously examined mediational links that included job demands and resources. We found that a supportive and collaborative school climate was negatively related to ECSE teachers' job burnout and depressive symptoms indirectly through their perceived stress after controlling for other job demands and resources. The current study is the first attempt to explore whether ECE and ECSE teachers' perceived stress mediates the links between job demands and resources and teachers' professional and psychological well-being. These mediation links are reasonable assumptions even though these are correlational links rather than causal links. According to Iacovides et al. (2003), job burnout and depression may be caused by perceived stress that may be related to individuals' internal characteristics as well as their social environments. However, we did not find this link among ECE teachers. Further investigation is necessary to more clearly understand the role of school/program climate in ECE teachers' well-being and its mediational links to ECSE teachers' well-being.

Fourth, regarding the role of teachers' internal resources, those who had stronger job commitment experienced less job burnout for both ECE and ECSE teachers. As Hakanen et al. (2006) have also mentioned, our findings show that ECE and ECSE teachers' job commitment is a

promising area in which effort should be focused to reduce teachers' job burnout while enhancing their well-being. Even though teachers may be exposed to similar stressors and to a similar school climate, individuals may or may not experience job burnout or suffer from depression depending on their own internal characteristics. Even though some researchers have viewed teachers' job commitment as part of professional well-being (e.g., Buettner et al., 2016; Hakanen et al., 2006; Kwon et al., 2021), there are few studies that have examined teachers' job commitment as a predictor of their psychological and professional well-being. Our findings are similar to those of earlier studies which examined job commitment as a predictor of teachers' well-being. For example, British general education teachers' job commitment predicted their perceived stress (Jepson & Forrest, 2006), and Chinese general education teachers' job commitment predicted their professional well-being (McInerney et al., 2015).

Fifth, we also found another possible internal resource for teachers' well-being — a sense of their teaching efficacy, which refers to a teacher's beliefs about their ability to work with and help their students. We found that ECE teachers' teaching efficacy was negatively associated with their perceived stress and job burnout, whereas ECSE teachers' teaching efficacy did not predict their perceived stress, job burnout, or depressive symptoms. Our findings among ECE teachers are consistent with those of other studies, including that of a meta-analysis investigation, all of which indicated that general education teachers' self-efficacy has a strong negative link with their perceived stress (Zee & Koomen, 2016) and job burnout (e.g., Ahola et al., 2014; Boujut et al., 2017; Browers & Tomic, 2000; Martin et al. 2012; Schwarzer & Hallum, 2008; Skaalvik & Skaalvik, 2007). The same is true for ECE teachers (Jeon et al., 2018). Unlike Jeon et al.'s study, the current study did not find a direct link between ECE teachers' teaching efficacy and their depressive symptoms. Instead, we found that ECE teachers' teaching efficacy was negatively linked to their job burnout both directly and indirectly through their perceived stress. Also, ECE teachers' teaching efficacy was negatively linked to their depressive symptoms through their perceived stress, but only indirectly. This means that ECE teachers with a higher sense of teaching efficacy were less likely to have perceived stress, and, in turn, were likely to experience less job burnout and fewer depressive symptoms. Improving ECE teachers' teaching efficacy may serve as a resource to reduce their perceived stress, job burnout, and depressive symptoms. Our study is one of the first attempts to examine ECE teachers' internal beliefs and psychological well-being by using a mediational model. Further investigation of the mediational relationships between ECE teachers' internal beliefs and psychological well-being is warranted.

Unlike ECE teachers, ECSE teachers' teaching efficacy was not related to their perceived stress, burnout, or depressive symptoms. The absence of a relationship between teaching efficacy and well-being among ECSE teachers is inconsistent with the findings of other studies. For example, Ruble et al. (2011) found a negative relationship between classroom management efficacy and job burnout among teachers of students with autism; however, Shyman (2010) did not find this link among special education paraeducators after controlling for job-demands and support, which is consistent with our finding. Inconsistencies in these findings may be attributed to the complex relationships between job-demands and resources, teaching efficacy, and well-being among ECSE teachers (Hopman et al., 2018). Hopman et al. found moderating relationships among job demands, teaching efficacy, and job burnout for Dutch special education teachers of students with psychiatric disabilities. For example, for ECSE teachers below a certain threshold of job-demands, teaching efficacy alone may help them reduce their perceived stress and job burnout. Additionally, there may be no link between teaching efficacy and ECSE teachers' well-being after controlling for other job demands and resources, since they are more prepared to teach children with disabilities because of their pre-service training and additional professional development. Further investigation is warranted to gain a better understanding of ECSE

teachers' teaching efficacy and their well-being.

4.1. Limitations

The current study predicted ECE and ECSE teachers' perceived stress, job burnout, and depressive symptoms by using a mediational model; the findings of the current study provide a better understanding of ECE and ECSE teachers' psychological and professional well-being and implications for their professional development and support programs. However, we acknowledge several limitations that would lead to caution in the interpretation of our findings. First, we examined a mediational model based on the previous literature, but our study is a correlational study. Therefore, we cannot determine the causality of the relationships between the constructs. Second, even though we carefully designed our sampling method to recruit representative study samples, there is selection bias in the teachers who volunteered to participate in our study. This is evident in the demographic characteristics of our ECE and ECSE teachers, where Caucasian teachers tended to participate in the study more than others. It is also possible that those who had seriously perceived stress and more experienced job burnout and depressive symptoms might exclude themselves from the study. Third, our ECE and ECSE teacher samples were limited by geographic location, and our findings are only generalizable to teachers in the United States. Fourth, we also compared ECE teachers' perceived stress, job burnout, and depressive symptoms to those of ECSE teachers. However, these two groups of teachers are different populations in terms of their educational qualification requirements and work conditions. The measures used to capture teachers' job stress, demands, and teaching efficacy were developed for general ECE settings and not for ECSE teachers. Further development of measurements is necessary to more accurately capture information about ECSE teachers and their settings.

4.2. Implications for practice and research

The current study findings suggest some job demands (e.g., job stressors), school climate, and internal resources (e.g., job commitment) may increase or prevent the likelihood of perceived stress, job burnout, and depressive symptoms for both ECE and ECSE teachers. The positive links between job stressors and negative psychological and professional well-being indicate the importance of reducing job-related stressors and increasing support and resources to alleviate psychological distress and improve well-being at multiple levels. For example, teachers need to learn the warning signs and symptoms of negative psychological and professional well-being, plan for and engage in self-care, and develop a repertoire of relaxation techniques (Kwon et al., 2020). Empowering and strengthening teachers' job commitment may also help teachers reduce negative psychological and professional well-being. Organizational-level support and resources are required, such as creating a positive and collaborative school climate, the ongoing training and support of leadership or administration, and providing peer support programs to teachers (especially for ECSE teachers). In particular, given our finding that ECE teachers who have more children with disabilities in the classrooms do feel more distress, it is important to provide additional staff, professional development opportunities, and coaching and mentoring programs for this population. Also, holistic mental health consultation, behavioral consultation, coaching programs and resources should be readily available in the local community and online.

5. Compliance with ethical standards

Human Participants and/or Animals Research involved human subjects, with all procedures approved by the University's Institutional Review Board.

Informed Consent Informed teacher consents were obtained for all study participants.

6. Author note

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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